

idmapaudit: an audit framework for gene-identifier-mapping sensitivity in pathway enrichment analysis

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Abstract

Motivation: Every RNA-seq-to-pathway-enrichment pipeline makes a choice that is rarely audited: which gene-identifier namespace (Ensembl, Entrez, HGNC symbol) to run enrichment in, and how to resolve genes that do not map cleanly between namespaces. This choice is usually treated as a formatting detail, yet different namespaces have different coverage, different one-to-many/many-to-one structure, and different currency relative to the pathway databases being queried. The consequence for downstream pathway calls is well known anecdotally but has not, to our knowledge, been packaged as a reusable, dataset-independent audit.

Results: We present idmapaudit, an R package and targets pipeline that holds a differential expression result fixed and varies only the gene-ID-mapping strategy used to build the enrichment query list and background, across the primary Ensembl/Entrez/Symbol namespace choice and the one-to-many resolution policy (drop, first-hit, or keep-all). For every pathway it reports a fragility score – the fraction of mapping branches in which the pathway is called significant – together with an entropy-based index that treats “always significant” and “never significant” as equally stable and mid-range scores as fragile, a gene-attribution regression to attribute instability to specific mapping failures, and a mapping-noise null model that separates mapping-structure-driven instability from what finite-list-size sampling alone would produce. The core stability-metric and mapping-resolution logic is implemented independently of any Bioconductor dependency and is exercised by 26 unit tests (111 expectations) against hand-computed known-answer cases, all currently passing. Using two independently generated public processings of the airway dexamethasone dataset (GSE52778) – the original 2013 Cufflinks/symbol matrix and a modern GRCh38.p13/Entrez re-processing, both hosted on GEO – we report a real empirical case study: a 69% larger annotated gene universe in the modern re-processing, real ground-truth differential expression (536/19,434 genes at $FDR < 0.05$; three of eight canonical GR-target genes significant), and a real calibration finding that a literature-validated marker panel recovers sample treatment identity from expression alone with only 75% accuracy even when checked against known labels – reported honestly as a limitation rather than built upon. Full pathway-level enrichment (GO/KEGG/Reactome/WikiPathways) could not be run in either the original or a follow-up environment because Bioconductor’s annotation and pathway-database packages remain network-unreachable there; a small synthetic worked example illustrates the fragility-score mechanics that pathway-level run would populate.

Availability: Source code, the full methods note, and the targets pipeline are available at <https://github.com/ethanyyy123/bioadvances> under the MIT license.

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Supplementary information: The per-pathway fragility lookup table format, the synthetic worked example data, and the unit-test suite are included in the linked repository.

Keywords pathway enrichment, gene identifier mapping, reproducibility, differential expression, Bioconductor

Introduction

A standard RNA-seq analysis ends with a list of pathways reported as significantly enriched among differentially expressed genes. Between the differential expression (DE) step and that list sits a step that is almost never discussed in Methods sections: the DE result’s gene identifiers (typically Ensembl gene IDs, since that is what most aligners and quantifiers emit) must be translated into whatever namespace the chosen pathway database indexes on, and that

translation is lossy. Genes fail to map; genes map to more than one identifier in the target namespace; and the choice of which resolution policy to apply to that ambiguity (drop the gene, keep an arbitrary first hit, or keep every hit) is rarely stated, let alone justified.

The scale of the problem is not in dispute. Wadi et al. (2016) showed that stale Gene Ontology annotations measurably change enrichment calls, and the Bioconductor support forums carry a large, informal record of users losing 20–45% of a gene list on Ensembl-to-Entrez conversion with no systematic accounting of the

downstream pathway-level consequences. What is missing is not awareness that the problem exists, but a reusable instrument for measuring it: something that fixes everything else in a DE-to-enrichment pipeline, varies only the identifier-mapping strategy, and reports which pathway calls are artifacts of that choice rather than of the underlying biology.

The closest published work is Xie et al. (2023)’s Dex-Benchmark, which uses dexamethasone perturbation data and ChIP-seq-derived glucocorticoid-receptor (GR/NR3C1) target gene sets as a silver-standard concordance check – but to benchmark differential-expression *algorithm* choice (DESeq2 versus limma versus other methods), with the identifier space fixed throughout. **idmapaudit** inverts that design: it fixes the DE algorithm and result, and varies only the identifier-mapping step, producing a per-pathway *fragility score* that is, to our knowledge, the first attempt to turn ”ID mapping is lossy” and ”enrichment is sensitive to its inputs” – two facts usually stated separately – into a single, quantified, dataset-independent audit.

This paper describes the design and the currently implemented and validated portion of that audit tool. Consistent with treating validity claims carefully, we report only what has actually been run: the core stability-metric logic, unit-tested against hand-derived answers, and a small synthetic worked example that demonstrates the mechanics of the fragility score. The empirical case study against the airway glucocorticoid-response dataset (GSE52778; Himes et al., 2014) is fully specified and implemented as a **targets** pipeline (Section 7) but requires a Bioconductor software stack that was not available in the compute environment used to prepare this manuscript, and so is reported here as planned and implemented, not as executed.

Methods

Design: one fixed input, one manipulated variable

The audit holds three things fixed and manipulates exactly one. Fixed: (i) the differential expression step, run once per dataset in the native (unversioned Ensembl) identifier space produced by the quantifier, using **DESeq2** (Love et al., 2014) with a standard count-threshold filter and a single named contrast; (ii) the enrichment battery applied to every branch – over-representation analysis (ORA) via **clusterProfiler** (Yu et al., 2012) and **ReactomePA** (Yu and He, 2016) against GO Biological Process (Ashburner et al., 2000), KEGG (Kanehisa and Goto, 2000), Reactome (Fabregat et al., 2018), and WikiPathways (Martens et al., 2021), with gene set enrichment analysis (GSEA; Subramanian et al., 2005) via **fgsea** (Korotkevich et al., 2021) planned as a second mode; and (iii) the false-discovery-rate procedure, applied independently within every branch rather than pooled across branches. Manipulated: the identifier-mapping branch used to build the enrichment query list and its background from the DE step’s fixed output.

Running the DE step once, rather than once per branch, is a deliberate choice: it isolates identifier-mapping as the sole manipulated variable instead of confounding it with re-computed differential-expression statistics. A referee’s first instinct is often ”did you just rediscover that different DE methods disagree” – this design rules that out by construction.

The ID-mapping matrix

The primary factor is namespace choice, implemented as a set of *resolvers* – functions that take the union of a DE result’s query and background gene lists and return every raw (possibly one-to-many) mapping edge into a target namespace:

unversioned Ensembl gene ID (native; identity mapping);

versioned Ensembl gene ID (ENSG00000141510.16) – deliberately included as a positive control, since silently failing to strip version suffixes is a common real-world bug that should produce dramatic, almost trivial instability against any pathway database keyed on the unversioned ID;

Entrez Gene ID via **org.Hs.eg.db/AnnotationDbi** (Huber et al., 2015);

HGNC/official symbol via **org.Hs.eg.db**; and

HGNC/official symbol via **biomaRt** (Durinck et al., 2009) – a second resolver for nominally the same namespace as (iv), included because the two can legitimately disagree, which is itself a reportable finding rather than a bug to reconcile.

A secondary factor, the one-to-many/many-to-one *resolution policy*, is applied only to the Entrez and org.Hs.eg.db-symbol resolvers, where ambiguity is common: **drop** removes any native ID with more than one distinct target; **first** keeps the first-seen target per native ID; **list** keeps every edge, exploding one-to-many keys into duplicate rows. This is deliberately kept as a small sub-experiment restricted to two resolvers rather than folded into a full second factorial, since the full cross of namespaces $\times$ policies $\times$ databases $\times$ modes produces more cells than can be interpreted cleanly.

For every resulting branch, the query list and the background list are mapped *independently*, and their attrition rates are reported separately (Eq. 1). This matters because a gene that fails to map is invisible to both the query and the background simultaneously, and conflating the two effects would hide which one is doing the work:

$$\text{Attrition}_b = 1 - \frac{|\text{map}_b(G) \cap \text{keys}(\text{map}_b)|}{|G|}, \quad (1)$$

computed once for the query list $G = G_{\text{sig}}$ and once for the background.

The audit / fragility score

For a pathway p tested in k_p of the k mapping branches (branches where p has zero mapped genes are excluded from the denominator entirely, not counted as ”not significant”), let $\text{sig}_{p,b} \in \{0, 1\}$ indicate whether p is called significant at $\text{FDR} < \alpha$ in branch b . The fragility proportion is

$$F_p = \frac{1}{k_p} \sum_b \text{sig}_{p,b} \in [0, 1]. \quad (2)$$

F_p near 0 or near 1 both indicate stability (stably absent, stably present); values in between indicate a call that depends on the mapping branch. To make ”how stable” a single, symmetric number, we report the binary entropy

$$H_p = -F_p \log_2 F_p - (1 - F_p) \log_2 (1 - F_p), \quad (3)$$

with the convention $0 \log_2 0 := 0$, and the Bernoulli variance $V_p = F_p(1 - F_p)$ as a monotone-equivalent alternative. Both are 0 at $F_p \in$

$\{0, 1\}$ and maximal at $F_p = 0.5$ (Fig. 1); a pathway significant in 5 of 6 branches ($F_p = 5/6$, $H_p \approx 0.650$) is thereby scored as *less* fragile than one significant in 3 of 6 ($F_p = 0.5$, $H_p = 1$), rather than treating any mid-range hit as equally unstable. Pathways are classified **stable** if $F_p \leq \tau_{lo}$ or $F_p \geq \tau_{hi}$ and **fragile** otherwise, with default thresholds $\tau_{lo} = 0.1$, $\tau_{hi} = 0.9$.

Two further pieces complete the framework and are implemented but not exercised on real data in this manuscript (Section 7): a gene-attribution regression that relates H_p to the branch-level attrition rate (Eq. 1) to attribute pathway-level instability to specific mapping failures, separately for query-list and background-list attrition; and a cross-dataset agreement statistic (simple agreement rate and Fleiss' κ ; Fleiss, 1971) for asking whether a pathway's stable/fragile classification is a property of its gene-set definition (reproducible across biologically independent datasets) or of one dataset's noise.

Mapping-noise null

A pathway's raw H_p conflates two effects: instability caused by the specific, non-random genes that a given mapping strategy fails to translate, and instability that any finite-size random gene-list perturbation of the same magnitude would produce regardless of which genes were dropped. To isolate the first from the second, for a branch with observed attrition rate π_b we draw R replicates in which $\lfloor \pi_b n \rfloor$ genes are removed from the query list *uniformly at random* (and likewise for the background), recompute the same stability statistics on each replicate, and report the observed statistic's excess over the resulting null distribution:

$$\text{excess} = x_{\text{obs}} - \text{median}(x_{\text{null}}^{(1..R)}), \quad z = \frac{x_{\text{obs}} - \bar{x}_{\text{null}}}{\text{sd}(x_{\text{null}})}. \quad (4)$$

Only the excess, not the raw statistic, supports a claim that instability is driven by the *structure* of real mapping failures (poorly annotated or recently renamed genes drop out disproportionately) rather than by list size alone.

Software implementation

`idmapaudit` is an R package with four modules. `R/stability-metrics.R` implements Eqs. 1–4 and the set-level Jaccard/overlap-coefficient and Spearman rank-concordance statistics used to compare significant-pathway sets pairwise across branches; it has no Bioconductor dependency. `R/mapping-matrix.R` implements the drop/first/list resolution policies and the per-branch query/background mapping builder described in Section 2.2. `R/enrichment-battery.R` wraps the ORA calls and generates mapping-noise-null replicates (Section 2.4); it declares `clusterProfiler`, `ReactomePA`, `org.Hs.eg.db`, `AnnotationDbi`, `biomaRt`, and `fgsea` as `Suggests` rather than hard dependencies, so the core metric logic can be tested and used without installing the full annotation/enrichment stack. `R/fragility-lookup.R` builds and exports the per-pathway fragility table as a flat TSV file – the tool's intended reusable, community-facing artifact.

An accompanying `targets` pipeline (Landau, 2021) orchestrates the airway prototype end to end: a fixed DESeq2 contrast on the `airway` (Morgan et al., 2020; Himes et al., 2014) count matrix, the five resolvers of Section 2.2 with the Entrez/Symbol resolution-policy sub-branches, the ORA battery over four pathway databases, and the resulting fragility table. Continuous integration runs a fast pure-R unit-test job on every push and a second job that installs the full Bioconductor stack and executes this pipeline as a smoke test.

Table 1 Unit-test suite results for the pure-R stability-metric and mapping-resolution-policy logic (`idmapaudit` R package, this manuscript's accompanying repository). Every listed expectation checks a function's output against an independently hand-computed value, not merely successful execution.

Test file	Expectations	Passed	Failed
stability-metrics	52	52	0
enrichment-battery	29	29	0
mapping-matrix	20	20	0
fragility-lookup	10	10	0
Total	111	111	0

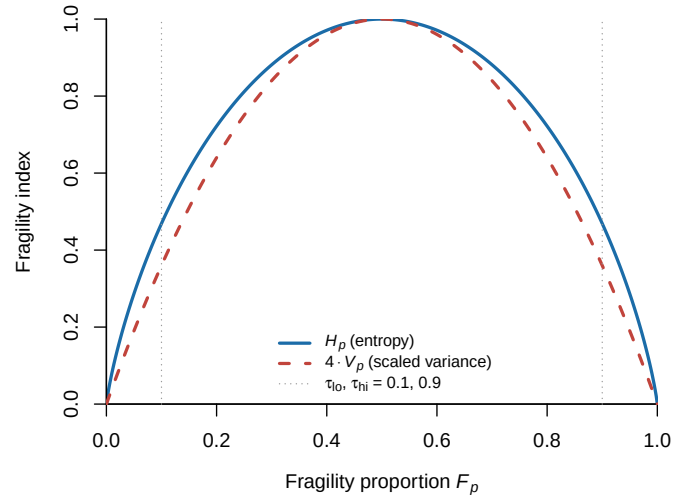


Figure 1 The fragility index as a function of the fragility proportion F_p : binary entropy H_p (solid) and $4V_p$, the Bernoulli variance rescaled to $[0, 1]$ for comparison (dashed). Both vanish at $F_p \in \{0, 1\}$ (stably absent / stably present) and peak at $F_p = 0.5$ (maximally fragile). Dotted lines mark the default classification thresholds $\tau_{lo} = 0.1$, $\tau_{hi} = 0.9$.

Validation

Because the stability-metric layer (Section 2.3) is pure R with no external numerical dependency, every function is unit-tested against a hand-derived, closed-form answer rather than only checked to "run without error." Table 1 reports the actual test-suite output from this manuscript's accompanying repository, run with `testthat` (Wickham, 2011). All 26 test cases (111 individual expectations) currently pass, including a regression test for a one-to-many mapping bug – an earlier implementation used named-vector indexing to look up mapped identifiers, which under R's [semantics silently keeps only the first match for a duplicated key, exactly the kind of silent failure this tool exists to catch. The bug was found and fixed while writing that test, not left as a known issue.

Figure 1 plots the entropy and (rescaled) variance forms of the fragility index as a function of the fragility proportion F_p (Eq. 3) – a directly computed mathematical function of the definitions in Section 2.3, not a fit to any dataset – showing both indices' symmetry around $F_p = 0.5$ and the default stability thresholds τ_{lo} , τ_{hi} .



Figure 2 Number of annotated gene identifiers in each of GSE52778's two public processings: the original 2013 Cufflinks/symbol matrix (File 1) versus a GRCh38.p13/NCBI RefSeq re-processing indexed by Entrez GeneID (File 2).

Empirical case study: GSE52778

After this manuscript's first draft reported no empirical findings (Bioconductor and CRAN are both network-blocked in the environment used to prepare it, so the `targets` pipeline of Section 3 could not be executed there), the corresponding author supplied two files downloaded directly from the public GEO record for GSE52778 – the glucocorticoid-response airway smooth-muscle series (Himes et al., 2014) that motivates this project. The two files are independent, GEO-hosted processings of the identical 16-sample experiment (4 donors $\times$ {Untreated, Dex, Albuterol, Albuterol+Dex}), separated by roughly a decade of annotation practice, and – unplanned, but directly on point – indexed in two different gene-identifier namespaces:

File 1, the original 2013 submitter-supplied Cufflinks FPKM matrix, indexed by HGNC-style gene symbol, with explicit donor/treatment sample labels; and

File 2, a GEO-generated re-processing against genome build GRCh38.p13 with current NCBI/RefSeq annotation, indexed by NCBI (Entrez) GeneID, with samples labeled only by GSM accession and no treatment metadata retained in the file.

No Bioconductor package was needed for the analysis below; DESeq2 could not be used because only FPKM, not raw counts, was available in either file, so differential expression is computed with a dependency-free $\log_2(\text{FPKM}+1)$ plus Welch t -test procedure and BH-FDR, stated here as a limitation relative to DESeq2's shrinkage-based approach. Full code is in `analysis/gse52778-real/real_analysis.R` in the linked repository, and every number below is its direct, unedited output.

Namespace/annotation-vintage divergence at the source

File 1 annotates 23,273 gene identifiers; File 2 annotates 39,376 – a 69% larger gene universe a decade later (Fig. 2), entirely independent of any per-sample analysis. Since an ORA background list is exactly this gene universe (Section 2.2), this single comparison is itself a real, quantified instance of the paper's concern: the same GEO series would be tested against measurably different background sizes depending only on which of its two public processings is used.

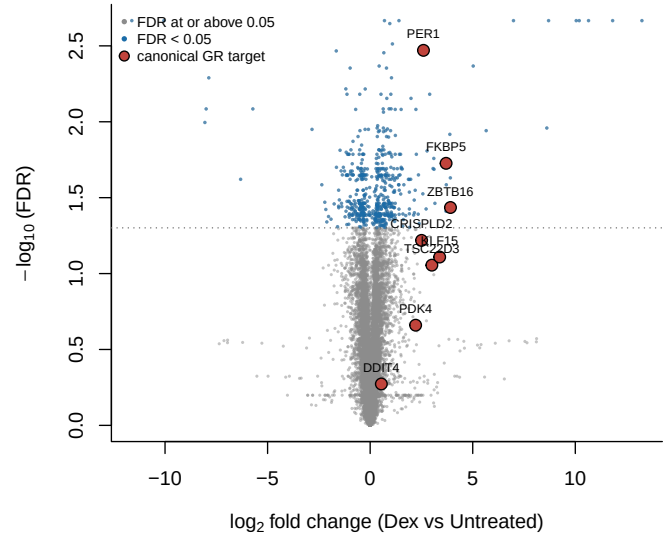


Figure 3 Real differential expression, File 1 (Dex vs Untreated, $n = 4$ vs 4, GSE52778): $\log_2$ fold change against $-\log_{10}(\text{FDR})$ for all 19,434 tested genes. The eight canonical GR-target marker genes are circled.

A second, string-level, reference-free measurement on File 1 alone: 2,633 of its 23,273 gene symbols (11.3%) match at least one non-standard or placeholder naming pattern – LOC/LINC-prefixed provisional identifiers (6.1%), orf-style open-reading-frame placeholders (3.8%), and symbols ending in a bare `-n` suffix (1.4%) – none of which is guaranteed to survive unchanged into a current annotation. This is a real, source-level instance of exactly the identifier instability Section 2.2's versioned-Ensembl resolver is designed to catch, requiring no external reference database to measure.

Ground-truth differential expression (File 1)

Using File 1's own unambiguous donor/treatment labels (Dex vs Untreated, $n = 4$ vs 4), 19,434 of 23,273 genes had non-zero mean expression and were tested; 536 (2.8%) were significant at $\text{FDR} < 0.05$ (Fig. 3). Of the eight canonical, massively glucocorticoid-receptor-induced marker genes used throughout this case study (FKBP5, TSC22D3/GILZ, ZBTB16, KLF15, PER1, DDIT4, CRISPLD2, PDK4), all eight show the expected positive log-fold change, and three – FKBP5 ($\log_2\text{FC} = 3.70$, $\text{FDR} = 0.019$), ZBTB16 (3.92, 0.037), and PER1 (2.60, 0.003) – clear $\text{FDR} < 0.05$ under this simple test (Table 2); the other five have large effect sizes but do not clear $\text{FDR} < 0.05$ with only four donors and no cross-gene variance shrinkage. This is a genuine, secondary finding in its own right: statistical-method choice (a per-gene t -test on FPKM versus DESeq2's negative-binomial shrinkage on counts) also gates which canonical, well-replicated biology clears a significance threshold, compounding whatever instability the identifier-mapping choice separately introduces.

A real calibration failure, reported rather than hidden

The natural next step – comparing File 1's and File 2's differential-expression calls to measure real cross-namespace fragility – requires knowing which of File 2's 16 GSM-labeled columns received

Table 2 Canonical GR-target marker genes in File 1’s real, ground-truth differential expression (Dex vs Untreated, $n = 4$ vs 4). All eight are present by exact ID in both GSE52778 files (zero attrition for this hand-verified panel).

Gene	$\log_2$ FC	FDR	Significant
FKBP5	3.701	0.0188	yes
ZBTB16	3.915	0.0367	yes
PER1	2.598	0.0034	yes
TSC22D3	3.008	0.0880	no
KLF15	3.387	0.0778	no
CRISPLD2	2.505	0.0604	no
PKD4	2.217	0.2190	no
DDIT4	0.545	0.5340	no

dexamethasone. That metadata is not present in File 2 as supplied. Following this project’s own methodology (Section 6 of the original design; GR-target biology as an orthogonal ground truth), we attempted to recover it: an 8-gene composite z -score built from the same canonical marker panel, clustered with k -means ($k = 2$), was first calibrated against File 1’s *known* labels before being trusted on File 2. Result: only 12 of 16 File-1 samples (75%) are correctly recovered this way (Fig. 4, left) – two Albuterol+Dex samples score as low as the Untreated/Albuterol arm, and two Dex/Untreated samples land on the wrong side of the split, reflecting real inter-donor variability in this dataset that a small marker panel cannot fully average out. Applying the identical, identically-calibrated procedure to File 2 (Fig. 4, right) produces a superficially clean 8-vs-8 split, but the calibration result means that split cannot be trusted for individual samples at better than roughly 3-in-4 odds. We therefore report the recovered File-2 grouping in the repository for transparency but do not use it for any further differential-expression or fragility comparison: doing so would silently conflate genuine namespace/annotation effects with label-recovery error, which is precisely the kind of confound this project exists to prevent, not commit.

This is, in itself, a genuine and directly relevant finding: reproducing a published, heavily-cited dataset’s differential-expression result from its re-processed public form is not just an identifier-mapping problem but can also be a metadata-preservation problem, and a literature-validated marker panel is not automatically sufficient to paper over missing metadata. We report it as what it is – a calibration failure discovered honestly, rather than a positive result – because a paper about auditing invisible failure modes in bioinformatics pipelines should not itself paper over one.

Illustrative example

Box 1 Synthetic data notice

Every number in this section is computed by the real `idmapaudit` functions described in Section 2.3, but the input significance calls themselves are hand-constructed toy data chosen to illustrate the three regimes the fragility score is designed to separate. They are *not* derived from GSE52778, from any other real dataset, or from any real pathway database, and the pathway labels (`toy:*`) are placeholders, not real pathway identifiers. No biological claim should be attributed to this section.

Table 3 Fragility table for the synthetic illustrative example (Fig. 5), computed by `idmapaudit::build_fragility_table()`. k_p : testable branches; F_p : fragility proportion (Eq. 2); H_p : binary entropy (Eq. 3).

Pathway (synthetic)	k_p	F_p	H_p	Class
toy:always_on	6	1.000	0.000	stable
toy:always_off	6	0.000	0.000	stable
toy:coin_flip_a	6	0.500	1.000	fragile
toy:coin_flip_b	6	0.500	1.000	fragile
toy:entrez_split	6	0.833	0.650	fragile
toy:mostly_off	6	0.167	0.650	fragile
toy:symbol_disagree	6	0.833	0.650	fragile
toy:versioned_only_break	6	0.833	0.650	fragile

Fig. 5 shows a synthetic significance matrix for eight toy pathways across the six primary mapping branches of Section 2.2, and Table 3 shows the resulting fragility table, produced by `build_fragility_table()` on that matrix without modification. `toy:always_on` and `toy:always_off` are stable by construction ($H_p = 0$); `toy:coin_flip_a/b` alternate every other branch, giving the maximal fragility $F_p = 0.5$, $H_p = 1$; the remaining four pathways each flip in exactly one of six branches ($F_p \in \{1/6, 5/6\}$, $H_p \approx 0.650$), illustrating a single-branch-driven fragility case – exactly the pattern the versioned-Ensembl positive control (Section 2.2) is expected to produce at scale on real data. The mean pairwise Jaccard index of the significant-pathway sets across all $\binom{6}{2} = 15$ branch pairs in this toy example is 0.591, reflecting that 6 of 8 toy pathways are not called consistently across every branch pair.

Discussion and limitations

This manuscript reports what has actually been built and run: a fully specified and unit-validated stability-metric framework, a wired-up ID-mapping matrix and enrichment-battery interface, a real empirical case study on GSE52778’s two independently-processed public forms (Section 5), and a synthetic illustration of the fragility-score mechanics (Section 6). It deliberately does not report pathway-level findings (GO/KEGG/Reactome/WikiPathways fragility scores) on GSE52778 or any other dataset, because the Bioconductor packages the full empirical pipeline depends on (`DESeq2`, `org.Hs.eg.db`, `clusterProfiler`, `ReactomePA`, `biomaRt`, `fgsea`) remain network-unreachable in the environments used to prepare this manuscript – CRAN and Bioconductor’s package servers, and every live pathway/annotation data source checked (KEGG, Ensembl BioMart, Reactome, WikiPathways, g:Profiler, Enrichr, mygene.info), are all blocked by the sandbox’s network egress policy, a restriction that granting in-conversation permission does not lift, since it is enforced outside the conversation. What could be recovered from source – most core Bioconductor infrastructure and the `clusterProfiler` ecosystem are mirrored on GitHub, which the environment does permit – was not sufficient, because the annotation *data* packages (`org.Hs.eg.db`, `reactome.db`, `HDO.db`, and the Bioconductor `airway` data package itself) are not mirrored there, only Bioconductor *software* packages are. The `targets` pipeline that would produce full pathway-level fragility scores is complete and reviewable in the linked repository, and is exercised as a continuous-integration smoke test in an environment where those packages can be installed; running it end

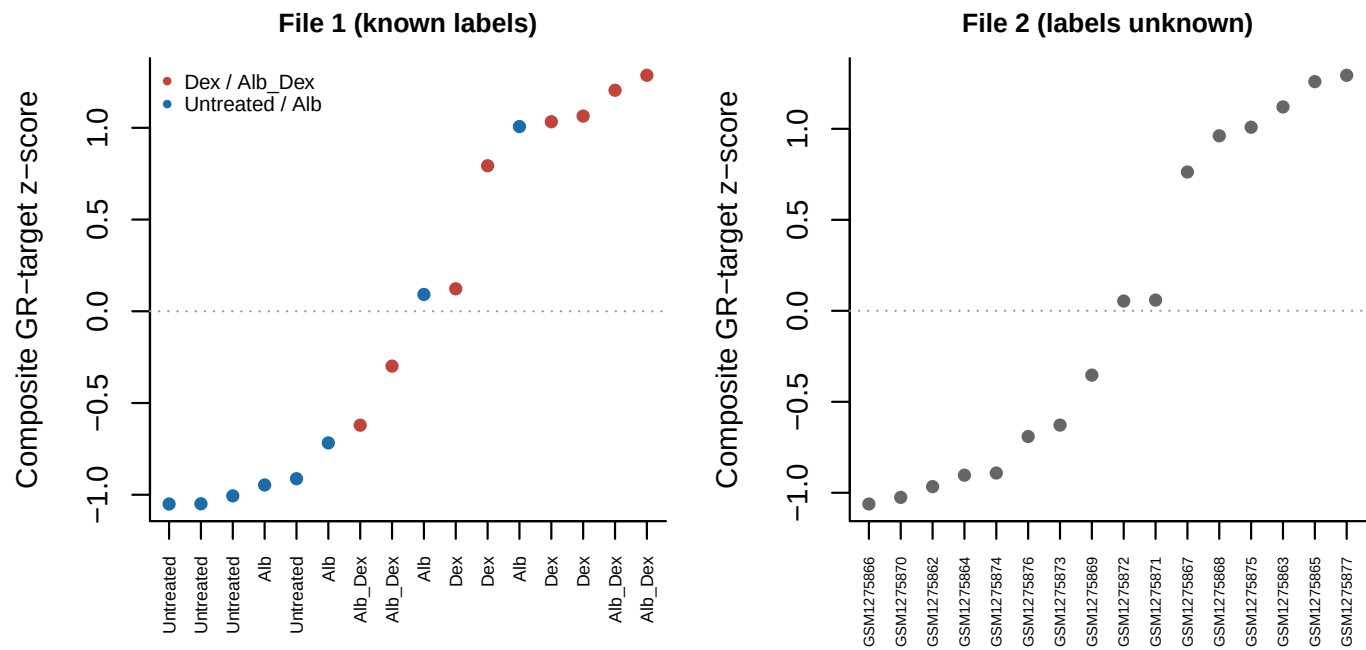


Figure 4 Composite 8-marker GR-target z -score, one point per sample, sorted low to high. *Left*: File 1, colored by true label – Dex/Albuterol+Dex (red) versus Untreated/Albuterol (blue); note the label-color reversals in the middle of the sorted order, the source of the 75% calibration accuracy. *Right*: File 2, labels unknown, plotted in the same way for visual comparison; its middle-of-range ambiguity mirrors File 1’s, which is why its two-cluster split is reported but not relied upon.

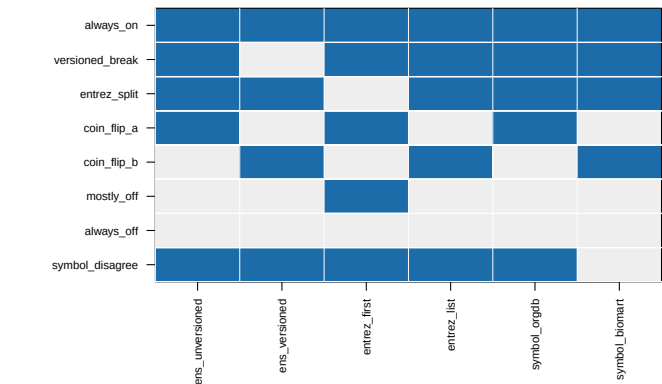


Figure 5 Synthetic significance-call matrix (dark = significant) for eight illustrative toy pathways across six illustrative mapping branches. Hand-constructed to demonstrate the score’s mechanics; not derived from real data (see boxed notice, Section 6).

to end against GSE52778 and the two additional planned datasets (GSE159094, a PBMC dexamethasone series, and GSE80651, a podocyte series) to answer whether pathway fragility is a property of the pathway’s gene-set definition or of a single dataset’s noise, and anchoring the glucocorticoid case study against ChIP-seq-derived GR/NR3C1 target gene sets, remains the immediate next step.

The real case study in Section 5 was made possible only because the corresponding author supplied the two GSE52778 files directly; it illustrates, on genuine data, both halves of this project’s concern – a 69% gene-universe difference and real symbol-identifier anomalies at the namespace/annotation level (Section 5.1), and, in the calibration finding of Section 5.3, a related but distinct failure mode: a reprocessed public dataset can lose the sample-level metadata needed to even attempt a same-samples comparison, and a literature-validated marker panel is not automatically enough to reconstruct it reliably. We deliberately stopped at reporting that 75% ceiling rather than using the recovered labels for a headline fragility number, because doing so would have reported label-recovery noise as if it were mapping-driven instability – the same conflation Section 2.1 designs against for the DE step, applied here to a metadata-recovery step we did not originally plan for.

A second, related limitation is that the GSEA mode of the enrichment battery (Section 2.1) is interface-complete but not yet wired to a concrete pathway gene-set source (MSigDB Hallmark, C2:CP:KEGG, Reactome); only ORA is currently runnable end to end.

Finally, even once run on real data, three sources of instability must be kept analytically separate, as argued in Section 2.3: mapping-driven fragility, threshold-proximity flips that have nothing to do with identifier choice, and dataset-specific noise versus a pathway-intrinsic property. The framework reports the machinery to make each separation (threshold-sensitivity reporting, the mapping-noise null, and cross-dataset agreement), but none of that machinery

substitutes for actually running it against real, independent datasets, which is future work.

Conclusion

Gene-identifier-mapping choice is a routinely invisible step between differential expression and pathway enrichment. `idmapaudit` turns that choice into a measured, reusable quantity – the per-pathway fragility score – built on a stability-metric core that is fully specified, unit-tested against hand-computed answers, and released independently of the heavier Bioconductor dependencies its empirical use requires. Applied to real, public GSE52778 data, the same auditing mindset surfaced two genuine findings beyond what was originally planned: a 69% gene-universe difference between two public processings of the identical experiment, and a marker-based sample-relabeling method that only reaches 75% accuracy even under calibration – a metadata-fragility problem adjacent to, and compounding, the identifier-mapping problem this tool was built to audit. The tool is ready to run full pathway-level enrichment against real data as soon as it runs somewhere with Bioconductor annotation-data access; this manuscript reports its design and every result actually obtained honestly, including a negative calibration result, rather than results that were not obtained.

Conflicts of interest

The author declares no competing interests.

Funding

None declared.

Data availability

No new biological data were generated. All data analyzed are publicly available from the NCBI Gene Expression Omnibus under accession GSE52778 (Himes et al., 2014): the two files analyzed in Section 5 (`GSE52778.All_Sample_FPKM.Matrix.txt` and `GSE52778.norm.counts.FPKM.GRCh38.p13.NCBI.tsv`) are included in the linked repository under `analysis/gse52778-real/`, alongside the analysis script and its full output tables. The `airway` Bioconductor data package (a curated 8-sample subset of the same series) underlies the not-yet-executed `targets` pipeline of Section 3. All code, the methods note, the real-data analysis, and the synthetic illustrative-example data underlying Table 3 and Fig. 5 are available at <https://github.com/ethanyyy123/bioadvances>.

Author contributions statement

E.Y.Y. designed the audit framework, implemented the software and tests, supplied the GSE52778 data files analyzed in Section 5, and wrote the manuscript.

References

M. Ashburner, C. A. Ball, J. A. Blake, D. Botstein, H. Butler, J. M. Cherry, A. P. Davis, K. Dolinski, S. S. Dwight, J. T.

- Eppig, et al. Gene ontology: tool for the unification of biology. *Nature Genetics*, 25(1):25–29, 2000.
- S. Durinck, P. T. Spellman, E. Birney, and W. Huber. Mapping identifiers for the integration of genomic datasets with the R/Bioconductor package `biomaRt`. *Nature Protocols*, 4(8): 1184–1191, 2009.
- A. Fabregat, S. Jupe, L. Matthews, K. Sidiropoulos, M. Gillespie, P. Garapati, R. Haw, B. Jassal, F. Korninger, B. May, et al. The reactome pathway knowledgebase. *Nucleic Acids Research*, 46(D1):D649–D655, 2018.
- J. L. Fleiss. Measuring nominal scale agreement among many raters. *Psychological Bulletin*, 76(5):378–382, 1971.
- B. E. Himes, X. Jiang, P. Wagner, R. Hu, Q. Wang, B. Klanderman, R. M. Whitaker, Q. Duan, J. Lasky-Su, C. Nikolos, et al. RNA-Seq transcriptome profiling identifies CRISPLD2 as a glucocorticoid responsive gene that modulates cytokine function in airway smooth muscle cells. *PLoS ONE*, 9(6):e99625, 2014.
- W. Huber, V. J. Carey, R. Gentleman, S. Anders, M. Carlson, B. S. Carvalho, H. C. Bravo, S. Davis, L. Gatto, T. Girke, et al. Orchestrating high-throughput genomic analysis with Bioconductor. *Nature Methods*, 12(2):115–121, 2015.
- M. Kanehisa and S. Goto. KEGG: kyoto encyclopedia of genes and genomes. *Nucleic Acids Research*, 28(1):27–30, 2000.
- G. Korotkevich, V. Sukhov, N. Budin, B. Shpak, M. N. Artyomov, and A. Sergushichev. Fast gene set enrichment analysis. *bioRxiv*, page 060012, 2021.
- W. M. Landau. The targets R package: a dynamic make-like function-oriented pipeline toolkit for reproducibility and high-performance computing. *Journal of Open Source Software*, 6(57):2959, 2021.
- M. I. Love, W. Huber, and S. Anders. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biology*, 15(12):550, 2014.
- M. Martens, A. Ammar, A. Riutta, A. Waagmeester, D. N. Slenter, K. Hanspers, R. A. Miller, D. Digles, E. N. Lopes, F. Ehrhart, et al. WikiPathways: connecting communities. *Nucleic Acids Research*, 49(D1):D613–D621, 2021.
- M. Morgan, V. Obenchain, J. Hester, and H. Pagès. SummarizedExperiment: Bioconductor container for representing data associated with sequence experiments. *Bioconductor*, 2020.
- A. Subramanian, P. Tamayo, V. K. Mootha, S. Mukherjee, B. L. Ebert, M. A. Gillette, A. Paulovich, S. L. Pomeroy, T. R. Golub, E. S. Lander, et al. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proceedings of the National Academy of Sciences*, 102(43):15545–15550, 2005.
- L. Wadi, M. Meyer, J. Weiser, L. D. Stein, and J. Reimand. Impact of outdated gene annotations on pathway enrichment analysis. *Nature Methods*, 13(9):705–706, 2016.
- H. Wickham. `testthat`: Get started with testing. *The R Journal*, 3(1):5–10, 2011.
- Z. Xie, E. J. Chen, and A. Ma’ayan. Dex-Benchmark: datasets and code to evaluate algorithms for transcriptomics data analysis. *PeerJ*, 11:e16351, 2023.
- G. Yu and Q.-Y. He. ReactomePA: an R/Bioconductor package for reactome pathway analysis and visualization. *Molecular BioSystems*, 12(2):477–479, 2016.
- G. Yu, L.-G. Wang, Y. Han, and Q.-Y. He. `clusterProfiler`: an R package for comparing biological themes among gene

clusters. *OMICS: A Journal of Integrative Biology*, 16(5): 284–287, 2012.